# Haldane gap and the LSM theorem

Hyeongmuk Lim

October 26, 2026

## Contents

**1 Quantization of spin** **1**
1.1 Spin coherent states . . . . . 1
1.2 Spinor path integral . . . . . 2

**2 Spin chain models** **5**
2.1 Ferromagnetic Heisenberg model . . . . . 6
2.2 Antiferromagnetic Heisenberg model . . . . . 6

**3 Renormalization group analysis** **7**
3.1 SO(3) nonlinear sigma model . . . . . 7
3.2 Large $N$ limit . . . . . 8
3.3 The role of topological term . . . . . 10

**4 The Lieb-Schultz-Mattis theorem** **11**
4.1 The original LSM theorem . . . . . 11
4.2 Generalized LSM theorem . . . . . 13

# 1 Quantization of spin

## 1.1 Spin coherent states

We define the element $k_z \in \mathrm{SU}(2)/\mathrm{U}(1) \approx \mathbb{S}^2$ by

$$k_z = \frac{1}{(1+\bar{z}z)^{1/2}} \begin{pmatrix} 1 & -\bar{z} \\ z & 1 \end{pmatrix} \qquad (z \in \mathbb{C}). \tag{1}$$

It has a Gauss decomposition ($G_o = \bar{N}HN$)

$$k_z = e^{U_- z} e^{-U_3 \log(1+\bar{z}z)} e^{-U_+ \bar{z}} \tag{2}$$

where $U_* = \frac{1}{2}\sigma_*$ and $\sigma_\pm = \sigma_1 \pm i\sigma_2$. While the LHS of Eq. (2) belongs to SU(2), the individual factors in RHS are elements of $\mathrm{SU}(2)_{\mathbb{C}} = \mathrm{SL}(2,\mathbb{C})$. Given the spin $(2j+1)$ representation $D^{(j)}$ of SU(2), the decomposition (2) enables one to define the operator

$$W(\bar{z}, z) := D^{(j)}(k_z) = e^{D^{(j)}(U_-)z} e^{-D^{(j)}(U_3)\log(1+\bar{z}z)} e^{-D^{(j)}(U_+)\bar{z}}, \tag{3}$$

which generalizes the Weyl operator for the Heisenberg-Weyl group. We also use the usual notation $D^{(j)}(U_i) = \hat{J}_i$.

Let $|0\rangle$ be the highest weight state of $D^{(j)}$, and define the coherent states $|z\rangle$ labeled by $z \in \mathbb{C}$ as

$$W(\bar{z}, z)|0\rangle = \frac{1}{(1+\bar{z}z)^j} e^{\hat{J}_- z}|0\rangle = \frac{1}{(1+\bar{z}z)^j}|z\rangle. \tag{4}$$

We denote the same state as

$$|e\rangle := W(\bar{z}, z)|0\rangle = \frac{1}{(1+\bar{z}z)^j}|z\rangle, \quad z = \frac{e^1 + ie^2}{1+e^3}, \quad (e^1)^2 + (e^2)^2 + (e^3)^2 = 1, \tag{5}$$

where $e \in \mathbb{S}^2$ is the unit vector that corresponds to $z$ under the stereographic projection. In terms of the $\hat{J}_z$ eigenstates,

$$|z\rangle = \sum_{m=-j}^{j} c_m(z)|m\rangle \qquad c_m(z) = \left[\frac{(2j)!}{(j-m)!(j+m)!}\right]^{1/2} z^{j-m} \tag{6}$$

so that, using the notation $\langle\bar{z}| := |z\rangle^\dagger$, the overlap between two coherent states is given by

$$\langle\bar{z}|z'\rangle = (1+\bar{z}z')^{2j}, \qquad \langle e|e'\rangle = \left(\frac{1+e\cdot e'}{2}\right)^j \left(\frac{1+\bar{z}z'}{1+z\bar{z}'}\right)^j. \tag{7}$$

Furthermore, the coherent states satisfy the completeness relation

$$\int d(\bar{z}, z)\frac{1}{(1+|z|^2)^{2j}}|z\rangle\langle\bar{z}| = 1, \qquad d(\bar{z}, z) = \frac{d\bar{z}\wedge dz}{2i}\frac{2j+1}{\pi(1+|z|^2)^2}, \tag{8a}$$

$$\int de\,|e\rangle\langle e| = 1, \qquad de = \frac{2j+1}{4\pi}\sin\theta d\theta d\varphi. \tag{8b}$$

## 1.2 Spinor path integral

### 1.2.1 Coherent state path integral

$$\langle e''|T\exp\left[-i\int_{t'}^{t''} dt\,\hat{H}(t)\right]|e'\rangle$$

$$= \lim_{N\to\infty}\int de_N \cdots \int de_1 \prod_{n=0}^{N}\langle e_{n+1}|e_n\rangle \exp\left[-i\Delta t_n H(e_{n+1}, e_n; t_n)\right] \tag{9}$$

where the classical Hamiltonian is given by

$$H(e_{n+1}, e_n; t_n) := \frac{\langle e_{n+1}|\hat{H}(t_n)|e_n\rangle}{\langle e_{n+1}|e_n\rangle}. \tag{10}$$

With $\Delta e_n := e_{n+1} - e_n$, we have

$$\langle e_{n+1}|e_n\rangle = 1 + \langle \Delta e_n|e_n\rangle \approx \exp\langle \Delta e_n|e_n\rangle. \tag{11}$$

This leads to

$$\begin{aligned}&\langle e''|T\exp\left[-i\int_{t'}^{t''} dt\,\hat{H}(t)\right]|e'\rangle \\ &= \lim_{N\to\infty}\int de_N\cdots\int de_1 \exp i\left[\sum_{n=0}^{N} -i\langle\Delta e_n|e_n\rangle - \Delta t_n H(e_{n+1},e_n;t_n)\right]\end{aligned} \tag{12}$$

In the continuum notation,

$$\langle e''|Te^{-i\int_{t'}^{t''} dt\,\hat{H}(t)}|e'\rangle = \int_{e'}^{e''} \mathcal{D}e\, e^{i\int_{t'}^{t''} dt\,\langle e|i\partial_t - \hat{H}|e\rangle}. \tag{13}$$

### 1.2.2 The Wess–Zumino term

To find the kinetic term of the classical Lagrangian $\mathcal{L} = \langle e|i\partial_t - \hat{H}|e\rangle$, we need to evaluate $\langle \Delta e_n|e_n\rangle$. To this end, note that (7) implies

$$\begin{aligned}\langle \Delta e|e\rangle &= \left(\frac{1+e\cdot e'}{2}\right)^j\left(\frac{1+\overline{z}'z}{1+z'\overline{z}}\right)^j - 1 \\ &= \left(1+\frac{\Delta e\cdot e}{2}\right)^j\left(1+\frac{z\Delta\overline{z}-\overline{z}\Delta z}{1+\overline{z}z}+\cdots\right)^j - 1 \\ &\approx j\cancel{\frac{1}{2}\Delta e\cdot e} + j\frac{z\Delta\overline{z}-\overline{z}\Delta z}{1+\overline{z}z} \\ &= -ji(1-\cos\theta)\Delta\varphi, \end{aligned} \tag{14a}$$

$$-i\langle de|e\rangle = -j(1-\cos\theta)d\varphi = -j\int_0^\theta \sin\theta' d\theta' d\varphi. \tag{14b}$$

The integral in (14b) represents the area of the infinitesimal triangle with vertices $e, e+de$, and $(0,0,1)$. Equivalently, we can extend the trajectory

$$e(t) = (\sin\theta(t)\cos\varphi(t), \sin\theta(t)\sin\varphi(t), \cos\theta(t)) \tag{15}$$

to a two-parameter function

$$\tilde{e}(s,t) = (\sin s\theta(t)\cos\varphi(t), \sin s\theta(t)\sin\varphi(t), \cos s\theta(t)) \tag{16}$$

so that the kinetic term for spin is given by the *Wess–Zumino term*,

$$\lim_{N\to\infty} -i\sum_{n=0}^{N}\langle\Delta e_n|e_n\rangle = j\int_{t'}^{t''} dt\int_0^1 ds\, e\cdot(\partial_t e\times\partial_s e) = j\cdot S_{\mathrm{WZ}}[e], \tag{17a}$$

$$S_{\mathrm{WZ}}[e] = \int_{t'}^{t''} dt\int_0^1 ds\,\tilde{e}\cdot(\partial_t\tilde{e}\times\partial_s\tilde{e}). \tag{17b}$$

In later sections, we will abbreviate $\tilde{e}$ simply by $e$.

### 1.2.3 Classical equation of motion

**Proposition 1.1.** *The variation of the Wess–Zumino term*

$$S_{\mathrm{WZ}}[e] = \int dt \int_0^1 ds\, e \cdot (\partial_t e \times \partial_s e) \tag{18}$$

*is given by*

$$\frac{\delta S_{\mathrm{WZ}}}{\delta e} = e \times \partial_t e. \tag{19}$$

*Proof.*

$$\begin{aligned}
&\delta[e \cdot (\partial_t e \times \partial_s e)] \\
&= \delta e \cdot (\partial_t e \times \partial_s e) + \partial_s \delta e \cdot (e \times \partial_t e) + \partial_t \delta e \cdot (\partial_s e \times e) \\
&= \delta e \cdot (\partial_t e \times \partial_s e) + \partial_s[\delta e \cdot (e \times \partial_t e)] - \delta e \cdot \partial_s (e \times \partial_t e) - \delta e \cdot \partial_t (\partial_s e \times e) + \partial_t(\cdots) \\
&= \partial_s[\delta e \cdot (e \times \partial_t e)] + \delta e \cdot [(\partial_t e \times \partial_s e) - \partial_s (e \times \partial_t e) - \partial_t (\partial_s e \times e)] + \partial_t(\cdots) \\
&= \partial_s[\delta e \cdot (e \times \partial_t e)] + 3\delta e \cdot (\partial_t e \times \partial_s e) + \partial_t(\cdots),
\end{aligned} \tag{20}$$

$$\begin{aligned}
&\int_0^1 ds\, \delta[e \cdot (\partial_t e \times \partial_s e)] \\
&= \delta e \cdot (e \times \partial_t e)\Big|_{s=0}^{s=1} + 3 \int_0^1 ds\, \delta e \cdot (\partial_t e \times \partial_s e) + \partial_t(\cdots)
\end{aligned} \tag{21}$$

Now, upon integration $\int dt$, the total derivative $\partial_t(\cdots)$ drops out. Furthermore, the three tangent vectors $\{\delta e, \partial_t e, \partial_s e\}$ of $\mathbb{S}^2$ are linearly dependent, hence the second term vanishes. Finally, $\delta e = 0$ on $s = 0$. This yields the desired result. □

**Proposition 1.2.** *Let $H(e)$ be a classical Hamiltonian for a single classical spin variable $e$, defined by $jH(e) = \langle e|\hat{H}|e\rangle$. Then the classical equation of motion is*

$$\dot{e} = \nabla_e H \times e. \tag{22}$$

*Proof.* We are looking for a stationary point for the action

$$S[e] = jS_{\mathrm{WZ}}[e] - j \int dt\, H(e). \tag{23}$$

By Proposition 1.1, the variation of the action is

$$\frac{1}{j}\frac{\delta S[e]}{\delta e} = e \times \dot{e} - \nabla_e H(e), \tag{24}$$

so we get an equation of motion $e \times \dot{e} = \nabla_e H$. Then using $e \cdot \dot{e} = 0$,

$$\dot{e} = (e \times \dot{e}) \times e = \nabla_e H \times e. \tag{25}$$

□

**Example 1.3.** *With spin angular momentum operators $\hat{J}_a$, consider the Hamiltonian*

$$\hat{H} = B \cdot \hat{J} = B^a \hat{J}_a \tag{26}$$

*where $B^a$ is interpreted as a magnetic field. Note that (26) is the most general form of Hamiltonian if $j = 1/2$. Since the classical Hamiltonian is*

$$H(e) = j^{-1}\langle e|\hat{H}|e\rangle = j^{-1}B^a\langle e|\hat{J}_a|e\rangle = B^a e_a, \tag{27}$$

*the classical motion is the familiar precession.*

$$\dot{e} = B \times e \tag{28}$$

# 2 Spin chain models

In this section, we analyze the Heisenberg model in one dimension (1D) semiclassically, using the spinor path integral established in Sec. 1.2. The quantum and classical Hamiltonians are given by

$$\hat{H} = \mp\frac{J}{2j}\sum_{\langle xx'\rangle}\hat{S}(x)\cdot\hat{S}(x') + \sum_x B^a\hat{S}_a(x) \qquad (J>0), \tag{29a}$$

$$H(e) = \mp\frac{J}{2j}\sum_{\langle xx'\rangle}S(x)\cdot S(x') + \sum_x B^a S_a(x) \qquad (J>0). \tag{29b}$$

where $\langle xx'\rangle$ runs over ordered pairs of adjacent lattice sites (each bond is counted twice). The minus (plus) sign before the coupling $J$ indicates that the model is ferromagnetic (antiferromagnetic). Together with the Wess–Zumino term, the action is given by

$$\frac{1}{j}S[e] = \int dt\left\{\int_0^1 ds\sum_x e(x)\cdot(\partial_t e(x)\times\partial_s e(x)) \pm \frac{J}{2}\sum_{\langle xx'\rangle}e(x)\cdot e(x') - \sum_x e(x)\cdot B\right\} \tag{30}$$

The continuum approximation of the coupling term can be obtained by using the Taylor expansion

$$e(x \pm a\hat{i}) = e(x) \pm a\nabla_i e(x) + \frac{1}{2}a^2\nabla_i^2 e(x) + O(a^3). \tag{31}$$

Integrating by parts and ignoring constant terms, the coupling term can be replaced by

$$\sum_{\langle xx'\rangle}e(x)\cdot e(x') = \sum_x\sum_i\sum_{x'=x\pm a\hat{i}} e(x)\cdot e(x') \mapsto -a^2\sum_x \nabla e(x)\cdot\nabla e(x). \tag{32}$$

The resulting continuum action is the integral of the Lagrangian density, $j^{-1}S[e] = \int dtdx\,\mathcal{L}[e]$, with

$$\mathcal{L}[e] = \int_0^1 ds\, e\cdot(\partial_t e\times\partial_s e) \mp \frac{J}{2}\nabla e\cdot\nabla e - e\cdot B \tag{33}$$

## 2.1 Ferromagnetic Heisenberg model

For a ferromagnet, we take the upper sign in front of the $J$-term in (33). Applying Proposition 1.2 to the Hamiltonian density (which can be read off from (33)),

$$\mathcal{H}[e] = \frac{J}{2}\nabla e \cdot \nabla e + e \cdot B, \tag{34}$$

we obtain an equation of motion

$$\dot{e} = Je \times \Delta e + B \times e. \tag{35}$$

Equation (35) is known as the *Landau–Lifshitz equation*.

## 2.2 Antiferromagnetic Heisenberg model

For an antiferromagnet, we take the lower sign in (33). Due to the positive coupling, one expects the ground state to have a Néel order where the spins are staggered. Anticipating this, we define a new variable $n(x)$ by

$$e(x) = (-1)^x n(x), \tag{36}$$

where $n(x) \in \mathbb{S}^2$ is supposed to be a smooth function of $x$ in the continuum limit. The coupling term changes sign upon rewriting the action in terms of the new variable $n(x)$:

$$\sum_{\langle xx' \rangle} e(x) \cdot e(x') = -\sum_{\langle xx' \rangle} n(x) \cdot n(x') \tag{37}$$

This flips the sign of the coupling term back to negative. Therefore, it is the Wess–Zumino term that distinguishes the behavior of antiferromagnet from ferromagnet:

$$\sum_x e(x) \cdot (\partial_t e(x) \times \partial_s e(x)) = \sum_x (-1)^x n(x) \cdot (\partial_t n(x) \times \partial_s n(x)). \tag{38}$$

A peculiar feature of (38) is that, because of the alternating sign, the Wess–Zumino terms of neighboring sites do not cancel: pairing neighboring sites leaves a gradient term, which is a topological term for the slowly varying field $n$. (Allowing a small uniform component of $e(x)$ in addition to the staggered one and integrating it out generates the kinetic term.) See [Gre13]. The (Euclidean) effective action for the slowly varying field is

$$S_{\text{eff}}[n] = \frac{1}{2g^2}\int d^2x\, \partial_\mu n \cdot \partial^\mu n + i\theta Q[n], \tag{39a}$$

$$Q[n] = \frac{1}{8\pi}\int d^2x\, \epsilon^{\mu\nu} n \cdot (\partial_\mu n \times \partial_\nu n), \tag{39b}$$

$$\theta = 2\pi j. \tag{39c}$$

(39c) indicates that the topological term has no effects for the integer spin case $j \in \mathbb{Z}$. However, as will be seen in Sec. 3.3, the topological term drastically changes the long wavelength behavior of the system for half-odd-integer $j$.

# 3 Renormalization group analysis

## 3.1 SO(3) nonlinear sigma model

In the Euclidean signature,

$$Z = \int \mathcal{D}\mathbf{n}\, \delta(\mathbf{n}(x)^2 - 1) \exp\left[-\frac{1}{2g^2}\int d^d x\, \nabla\mathbf{n}\cdot\nabla\mathbf{n}\right] \tag{40}$$

Decompose the field $n^a(x)$ into a slowly varying background $\tilde{n}^a(x)$ and a transversal fluctuation $\chi_\alpha(x)$:

$$n^a(x) = \tilde{n}^a(x)(1-\chi(x)^2)^{1/2} + \sum_{\alpha=1}^{N-1}\chi_\alpha(x)e^a_\alpha(x) \quad (\chi(x)^2 \ll 1) \tag{41}$$

We will expand the action in $\chi_\alpha(x)$ and only keep those terms up to quadratic order. First,

$$\begin{aligned}
\nabla n^a &= \nabla\tilde{n}^a(1-\chi^2)^{1/2} + \tilde{n}^a\nabla(1-\chi^2)^{1/2} + \nabla(\chi_\alpha e^a_\alpha) \\
&= \nabla\tilde{n}^a(1-\frac{1}{2}\chi^2) - \tilde{n}^a\chi_\alpha\nabla\chi_\alpha + (\nabla\chi_\alpha)e^a_\alpha + \chi_\alpha\nabla e^a_\alpha + O(\chi^2) \\
(\nabla n)^2 &= (\nabla\tilde{n})^2(1-\chi^2) + 2\nabla\tilde{n}^a\nabla(\chi_\alpha e^a_\alpha) + (\nabla\chi_\alpha)^2 \\
&\quad + \chi_\alpha\chi_\beta\nabla e^a_\alpha\nabla e^a_\beta + 2(\nabla\chi_\alpha)\chi_\beta e^a_\alpha\nabla e^a_\beta + O(\chi^3)
\end{aligned} \tag{42}$$

where we used $(\nabla\tilde{n}^a)\tilde{n}_a = 0$. The partition function is now

$$Z = \int \mathcal{D}\tilde{n}\,\delta(\tilde{n}^2-1)e^{-\frac{1}{2g^2}\int d^dx\,(\nabla\tilde{n})^2}\int\mathcal{D}\chi\, e^{-\frac{1}{2g^2}\int d^dx\,(\nabla\chi)^2}e^{-F_I[\tilde{n},\chi]}, \tag{43a}$$

$$F_I = \frac{1}{2g^2}\int d^dx\,\left[-\chi^2(\nabla\tilde{n})^2 + 2\nabla\tilde{n}^a\nabla(\chi_\alpha e^a_\alpha) + \chi_\alpha\chi_\beta\nabla e^a_\alpha\nabla e^a_\beta + 2(\nabla\chi_\alpha)\chi_\beta e^a_\alpha\nabla e^a_\beta\right]. \tag{43b}$$

Now, the contribution of the fluctuating field $\chi$,

$$\langle e^{-F_I}\rangle_\chi = \int\mathcal{D}\chi\, e^{-\frac{1}{2g^2}\int d^dx\,(\nabla\chi)^2}e^{-F_I[\tilde{n},\chi]} = 1 - \langle F_I\rangle_\chi + \frac{1}{2}\langle F_I^2\rangle_\chi - \cdots, \tag{44}$$

may contain kinetic terms $\sim(\nabla\tilde{n})^2$ and hence $g^2$ undergoes RG running. At leading order, we only need to keep $\langle F_I\rangle_\chi$:

$$\langle F_I\rangle_\chi = \frac{1}{2g^2}\int d^dx\,[-\delta_{\alpha\beta}(\nabla\tilde{n})^2 + \nabla e^a_\alpha\nabla e^a_\beta]\langle\chi_\alpha(x)\chi_\beta(x)\rangle_\chi \tag{45}$$

Using the relations

$$\langle\chi_\alpha(x)\chi_\beta(x)\rangle_\chi = g^2\delta_{\alpha\beta}\int_{\Lambda/\zeta}^{\Lambda}\frac{d^dq}{(2\pi)^d}\frac{1}{q^2} =: g^2\delta_{\alpha\beta}I_d, \tag{46a}$$

$$\begin{aligned}
\nabla e^a_\alpha\nabla e^a_\alpha &= \nabla e^a_\alpha\nabla e^b_\alpha(\tilde{n}^a\tilde{n}^b + e^a_\beta e^b_\beta) \\
&= e^a_\alpha e^b_\alpha(\nabla\tilde{n}^a)(\nabla\tilde{n}^b) + (e^a_\beta\nabla e^a_\alpha)(e^b_\beta\nabla e^b_\alpha) \\
&= (\nabla\tilde{n}^a)(\nabla\tilde{n}^a) + (e^a_\beta\nabla e^a_\alpha)^2,
\end{aligned} \tag{46b}$$

one has

$$\langle F_I \rangle_\chi = (-\delta_{\alpha\beta}\delta^{\alpha\beta} + 1) I_d \int d^d x \, \frac{1}{2} (\nabla \tilde{n})^2$$
$$= (2 - N) I_d \int d^d x \, \frac{1}{2} (\nabla \tilde{n})^2 \tag{47}$$

and after rescaling of all lengths by $1/\zeta$ (so that the momentum cutoff is set back to $\Lambda$),

$$\frac{1}{g^2(\Lambda/\zeta)} = \zeta^{d-2} \left[ \frac{1}{g^2(\Lambda)} + (2 - N) I_d \right] \tag{48}$$

From (48) we can compute the beta function of $g$. First, the integral $I_d$ is evaluated to be

$$I_d = \frac{\Omega_{d-1}}{(2\pi)^d} \Lambda^{d-2} \left[ \frac{\zeta^{2-d} - 1}{2 - d} \right], \qquad \Omega_{d-1} = \mathrm{Vol}(\mathbb{S}^{d-1}). \tag{49}$$

In particular, in $d = 2$ dimensions,

$$\frac{1}{g^2(\Lambda/\zeta)} = \frac{1}{g^2(\Lambda)} + (2 - N) \frac{1}{2\pi} \log \zeta, \tag{50}$$

$$\beta(g) = \frac{dg(\Lambda/\zeta)}{d \log \zeta} = (N - 2) \frac{g^3}{4\pi}. \tag{51}$$

Therefore, the beta function is positive for $N \geq 3$ and the coupling $g^2$ gets larger and larger as we flow to the infrared.

## 3.2 Large $N$ limit

Consider the SO($N$) model which extends the case $N = 3$.

$$Z = \int \mathcal{D}\mathbf{n} \mathcal{D}\sigma \, \exp \left[ -\frac{1}{2g^2} \int d^d x \, \{ \nabla n^a \nabla n_a + i\sigma (n^a n_a - 1) \} \right] \tag{52}$$

Integrate out the $n^a(x)$ fields to get an effective action for $\sigma(x)$:

$$Z = \int \mathcal{D}\mathbf{n} \mathcal{D}\sigma \, \exp \left[ - \int d^d x \, \frac{1}{2g^2} n^a (-\Delta + i\sigma) n_a \right] \exp \left[ \frac{i}{2g^2} \int d^d x \, \sigma \right]$$
$$\propto \int \mathcal{D}\sigma \, [\det(-\Delta + i\sigma)]^{-N/2} \exp \left[ \frac{i}{2g^2} \int d^d x \, \sigma \right]$$
$$= \int \mathcal{D}\sigma \, \exp \left[ -\frac{N}{2} \mathrm{Tr} \log(-\Delta + i\sigma) + \frac{i}{2g^2} \int d^d x \, \sigma \right] \tag{53a}$$

$$S_{\text{eff}}[\sigma] = \frac{N}{2} \mathrm{Tr} \log(-\Delta + i\sigma) - \frac{i}{2g^2} \int d^d x \, \sigma \tag{53b}$$

We will investigate the limit $N \to \infty$. In this limit, the majority of contributions of the path integral is concentrated near the saddle point in the function space of $\sigma(x)$. Denoting $G_\sigma(x, x') := (-\Delta + i\sigma(x))^{-1}$,

$$\frac{\delta}{\delta \epsilon} S_{\text{eff}}[\sigma + \epsilon \delta\sigma] = \frac{i}{2} N \, \mathrm{Tr}[G_\sigma \delta\sigma] - \frac{i}{2g^2} \int d^d x \, \delta\sigma(x) \tag{54a}$$

$$\frac{\delta^2}{\delta \epsilon^2} S_{\text{eff}}[\sigma + \epsilon \delta\sigma] = \frac{N}{4} \mathrm{Tr}[G_\sigma \delta\sigma G_\sigma \delta\sigma] \tag{54b}$$

For $\sigma = \sigma_0(x)$ to be a saddle, the Hessian $\mathrm{Tr}[G_\sigma \delta\sigma G_\sigma \delta\sigma]$ must be strictly positive definite. The first implication of this fact is that $\sigma_0$ must be translation invariant, i.e. constant in space: Otherwise, translation invariance of the effective action $S_{\text{eff}}$ would imply

$$S_{\text{eff}}[\sigma(\cdot + a)] = S_{\text{eff}}[\sigma(\cdot)] \quad \Rightarrow \quad \text{Hessian has a zero eigenvalue.} \tag{55}$$

The way out is to assume that the null vector $\delta_\epsilon \sigma = \sigma_0(\cdot + \epsilon) - \sigma_0(\cdot)$ is itself zero, i.e. $\sigma_0$ is invariant under translation. Thus, let us assume the constancy of $\sigma_0(x) \equiv -i\mu^2$. The Hessian can then be evaluated using the momentum bases.

$$\begin{aligned}
\text{Hess} &\propto \int \frac{dp}{(2\pi)^d}\frac{dk}{(2\pi)^d}\frac{dk'}{(2\pi)^d}\frac{dk''}{(2\pi)^d}\langle p|G_\sigma|k\rangle\langle k|\delta\sigma|k'\rangle\langle k'|G_\sigma|k''\rangle\langle k''|\delta\sigma|p\rangle \\
&= \int \frac{dp}{(2\pi)^d}\frac{dp'}{(2\pi)^d}\frac{1}{p^2+\mu^2}\frac{1}{(p')^2+\mu^2}\widehat{\delta\sigma}(p-p')\widehat{\delta\sigma}(p'-p) \\
&= \int \frac{dk}{(2\pi)^d}\underbrace{\int \frac{dp}{(2\pi)^d}\frac{1}{p^2+\mu^2}\frac{1}{(p+k)^2+\mu^2}}_{=:\Pi(k)}\widehat{\delta\sigma}(k)\widehat{\delta\sigma}(-k) \\
&= \int \frac{dk}{(2\pi)^d}\Pi(k)|\widehat{\delta\sigma}(k)|^2 > 0
\end{aligned} \tag{56}$$

where $\mu \in \mathbb{R}$ is implicit (see below) and we used

$$\langle p|G_\sigma|k\rangle = \frac{1}{p^2+\mu^2}(2\pi)^d\delta^{(d)}(p-k), \tag{57a}$$

$$\langle p|\delta\sigma|k\rangle = \int d^dx\, \delta\sigma(x)e^{-i(p-k)x} = \widehat{\delta\sigma}(p-k), \tag{57b}$$

$$\widehat{\delta\sigma}(-k) = \widehat{\delta\sigma}(k) \qquad \because \delta\sigma(x) \in \mathbb{R}. \tag{57c}$$

This shows that a constant $\sigma_0$ qualifies as a saddle point. Now, it remains to compute the value of $\mu^2$ by setting (54a) to zero:

$$\mathrm{Tr}\, G_{\sigma_0} = \int^\Lambda \frac{d^dk}{(2\pi)^d}\frac{1}{k^2+\mu^2} = \frac{1}{g^2N}. \tag{58}$$

In $d = 2$ dimensions, (58) leads to

$$\frac{1}{4\pi}\log\left(\frac{\Lambda^2+\mu^2}{\mu^2}\right) = \frac{1}{g^2N} \quad \Rightarrow \quad \mu^2 = \frac{\Lambda^2}{e^{4\pi/g^2N}-1}. \tag{59}$$

This justifies a posteriori the assumption that $\mu$ is real. Also, the result (59) indicates that the field $\mathbf{n}(x)$ remains massive ($\mu^2 > 0$) in the strongly coupled deep IR limit.

**Remark 3.1.** *Curiously, the saddle $\sigma_0(x) = -i\mu^2$ turns out to be imaginary, but there is no problem in (52) when we used*

$$\delta(\mathbf{n}^2-1) \propto \int \mathcal{D}\sigma\, \exp\left[-\frac{i}{2g^2}\sigma(\mathbf{n}^2-1)\right], \tag{60}$$

*because the only field which needs reality is $\delta\sigma$: Plugging $\sigma = -i\mu^2 + \delta\sigma$,*

$$\delta(\mathbf{n}^2-1) \propto \int \mathcal{D}\delta\sigma\, \exp\left[-\frac{\mu^2}{2g^2}(\mathbf{n}^2-1)\right]\exp\left[-\frac{i}{2g^2}\delta\sigma(\mathbf{n}^2-1)\right]. \tag{61}$$

### 3.3 The role of topological term

We observed in Sec. 2.2 that the antiferromagnetic (AF) Heisenberg chain with half integer spin acquires a topological term

$$Q[\mathbf{n}] = \frac{1}{8\pi}\int \epsilon_{abc} n^a dn^b \wedge dn^c \tag{62}$$

in the continuum limit.

We lift the field $\mathbf{n}(x) \in \mathbb{S}^2$ to a SU(2)-valued field $g(x)$, using the Hopf fibration $\mathrm{SU}(2) \to \mathbb{S}^2$.

$$n^a = \frac{1}{2}\operatorname{tr}(\sigma^a g \sigma^3 g^{-1}), \qquad g = \begin{pmatrix} z & -\bar{w} \\ w & \bar{z} \end{pmatrix} \in \mathrm{SU}(2). \tag{63}$$

Then the action functional can be rewritten in terms of the SU(2) field:

$$S[\mathbf{n}] = \frac{1}{2\lambda}\int d^2x\, \partial_\mu n^a \partial^\mu n^a + i\theta Q[\mathbf{n}], \qquad \theta = 2\pi j \tag{64}$$

$$\rightsquigarrow S[g] = \frac{1}{2\lambda}\int d^2x\, \operatorname{tr}(\partial_\mu g^{-1}\partial^\mu g) + ik\Gamma_{\mathrm{WZ}}[g], \qquad k = 2j. \tag{65}$$

where the kinetic term along the U(1) fiber direction is ignored since it can be absorbed in the current-current coupling $\mathbf{J}_L \cdot \mathbf{J}_R$, to be discussed below. In other words, we are embedding the observable algebra of the $\mathbb{S}^2$ model into that of the SU(2) sigma model (65). This identification is heuristic: since $\pi_2(\mathrm{SU}(2)) = 0$, a global lift $g$ exists only for configurations with $Q[\mathbf{n}] = 0$, so the $\theta$-term is not literally mapped to $k\Gamma_{\mathrm{WZ}}[g]$, and the two models need not have the same renormalization group flow. The WZW description below should be regarded as a physically motivated conjecture rather than a consequence of this lift.

For a particular value $\lambda = \lambda_*$ of the coupling strength, the model (65) coincides with the $\mathrm{SU}(2)_{k=2j}$ Wess–Zumino–Witten (WZW) conformal field theory. The deviation from the WZW fixed point is proportional to the current-current coupling:

$$S[g] = S_{\mathrm{SU}(2)_{2j}\,\mathrm{WZW}} + \lambda' \int d^2x\, \mathbf{J}_L \cdot \mathbf{J}_R, \qquad \lambda' \propto \lambda - \lambda_*. \tag{66}$$

It turns out that, for $k = 1$, $\mathbf{J}_L \cdot \mathbf{J}_R$ is the only SU(2)- and translation-invariant relevant or marginal scalar, hence the only deformation that potentially drives the AF Heisenberg chain out of the conformal fixed point. However, it is marginally irrelevant.

Moreover, for $k > 1$ odd, the $\mathrm{SU}(2)_k$ WZW model is not a stable fixed point—there are operators that destabilize it. The ultimate destination of the RG flow turns out to be the $\mathrm{SU}(2)_1$ WZW model, which is a stable fixed point. Thus, without fine tuning, the uniform AF Heisenberg chain (half-odd-integer $j$) flows to the $\mathrm{SU}(2)_1$ WZW CFT in the IR (central charge $c = 1$), with only logarithmic corrections from the marginally irrelevant $\mathbf{J}_L \cdot \mathbf{J}_R$. This finally reveals that the AF Heisenberg chain with half integer spin is gapless, in stark contrast with the integer spin case.

# 4 The Lieb-Schultz-Mattis theorem

We follow the functional-analytic approach of [Tas22]. The system of our interest is the quantum spin chain, by which we mean a tensor product Hilbert space and corresponding observable algebra

$$\mathcal{H} = \bigotimes_{n\in\mathbb{Z}} \mathbb{C}^{2j+1}, \qquad \mathfrak{A} = \bigotimes_{n\in\mathbb{Z}} \mathrm{Mat}_{2j+1,2j+1}(\mathbb{C}) \tag{67}$$

For each decomposable element $\hat{A} = \otimes_n A_n$ we define the *support* of $\hat{A}$ as

$$\mathrm{Supp}\,\hat{A} = \{\, n \in \mathbb{Z} \,|\, A_n \neq I \,\}. \tag{68}$$

For general elements, we define its support by the set of sites $x$ on which the operator acts nontrivially.

**Definition 4.1.** *A local operator is an observable $\hat{A} \in \mathfrak{A}$ with finite support.*

**Definition 4.2** (state)**.** *A state is a linear functional $\rho : \mathfrak{A}_{\mathrm{loc}} \to \mathbb{C}$ such that $\rho(\hat{1}) = 1$ and $\rho(\hat{A}^\dagger \hat{A}) \geq 0$ for any $\hat{A} \in \mathfrak{A}_{\mathrm{loc}}$.*

**Definition 4.3** (ground state)**.** *A ground state of $\hat{H}$ is a state $\omega$ such that*

$$\omega(\hat{V}^\dagger[\hat{H}, \hat{V}]) \geq 0 \tag{69}$$

*for any $\hat{V} \in \mathfrak{A}_{\mathrm{loc}}$. It follows from the definition that $|\rho(\hat{A})| \leq \|\hat{A}\|$ and $\rho(\hat{A}^\dagger) = \rho(\hat{A})^*$.*

**Definition 4.4** (locally unique gapped ground state)**.** *A ground state $\omega$ of $\hat{H}$ is called a locally unique gapped ground state if there exists a constant $\gamma > 0$ such that*

$$\omega(\hat{V}^\dagger[\hat{H}, \hat{V}]) \geq \gamma\omega(\hat{V}^\dagger\hat{V}) \tag{70}$$

*for any $\hat{V} \in \mathfrak{A}_{\mathrm{loc}}$ with $\omega(\hat{V}) = 0$. We call the largest such $\gamma$ the energy gap $\Delta E$.*

**Theorem 4.5.** *A locally unique gapped ground state is a pure split state.*

## 4.1 The original LSM theorem

We consider the Hamiltonians that are sum of the local terms $\hat{h}_j$, where each term satisfies

$$(\hat{U}_\theta^I)^\dagger \hat{h}_j \hat{U}_\theta^I = \hat{h}_j, \quad \hat{U}_\theta^I = \exp\left[-i\sum_{j\in I} \theta \hat{S}_j^{(z)}\right], \tag{71}$$

where $I$ is a finite interval that contains the support of $\hat{h}_j$.

**Example 4.6** (XXZ model)**.** *The XXZ model under uniform magnetic field is given by the Hamiltonian*

$$\hat{H}_{\mathrm{XXZ}} = \sum_{j\in\mathbb{Z}} J(\hat{S}_j^{(x)}\hat{S}_{j+1}^{(x)} + \hat{S}_j^{(y)}\hat{S}_{j+1}^{(y)}) + J'\hat{S}_j^{(z)}\hat{S}_{j+1}^{(z)} - H\hat{S}_j^{(z)}. \tag{72}$$

Given such a Hamiltonian and a pair $(x, l) \in \mathbb{R} \times \mathbb{R}_{>}$, we introduce the twist parameter $\theta_j$ for $j \in \mathbb{Z}$,

$$\theta_j = \begin{cases} 0 & (j < x), \\ 2\pi(j-x)/l & (x \le j < x+l), \\ 2\pi & (j \ge x+l), \end{cases} \tag{73}$$

and define the local twist operator

$$\hat{U}_{x,l} = \exp\left[-i \sum_{j \in \mathbb{Z}} \theta_j (\hat{S}_j^{(z)} + S)\right] \tag{74}$$

which is supported on the interval $[x, x+l] \cap \mathbb{Z}$. The following result is due to Lieb, Schultz, and Mattis [LSM61]:

**Lemma 4.7.** *Let $\omega$ be a ground state of an $\mathrm{U}(1)$ invariant Hamiltonian $\hat{H}$. Then there exist some constants $C$ and $l_0$, which depend on the Hamiltonian, that*

$$0 \le \omega(\hat{U}_{x,l}^{\dagger}[\hat{H}, \hat{U}_{x,l}]) \le \frac{C}{l} \tag{75}$$

*for all $x \in \mathbb{R}$ and $l \ge l_0$.*

*Proof.* See [Tas20]. □

The authors of [LSM61] were able to find low-lying states of the antiferromagnetic Heisenberg chain with spin 1/2. This proved that the system cannot have a unique gapped ground state (it is either gapless or has degenerate ground states), in contrast with the integer spin case where a unique ground state is separated from the other states by the Haldane gap.

Lemma 4.7 equally applies to the systems with locally unique gapped ground states, but implies different consequences. If $\omega$ is such a ground state, then the twisted state $\hat{A} \mapsto \omega(\hat{U}_{x,l}^{\dagger} \hat{A} \hat{U}_{x,l})$ must converge to the same state with $\omega$ in the thermodynamic limit. This is the content of the following:

**Lemma 4.8.** *Let $\omega$ be a locally unique gapped ground state with energy gap $\Delta E > 0$ of a $\mathrm{U}(1)$ invariant Hamiltonian $\hat{H}$. Then with the same constants $C$ and $l_0$ as in Lemma 4.7, we have*

$$|\omega(\hat{U}_{x,l})|^2 \ge 1 - \frac{C}{l} \frac{1}{\Delta E} \tag{76}$$

*for all $x \in \mathbb{R}$ and $l \ge l_0$.*

*Proof.* Set $\hat{W} = \hat{U}_{x,l} - \omega(\hat{U}_{x,l})$ so that $\omega(\hat{W}) = 0$. Using the defining property of locally unique gapped ground state,

$$\omega(\hat{W}^{\dagger}[\hat{H}, \hat{W}]) \ge \Delta E \omega(\hat{W}^{\dagger} \hat{W}). \tag{77}$$

Substituting the definition of $\hat{W}$,

$$\omega(\hat{U}_{x,l}^{\dagger}[\hat{H}, \hat{U}_{x,l}]) \ge \omega(\hat{U}_{x,l}) \omega([\hat{H}, \hat{U}_{x,l}]) + \Delta E (1 - |\omega(\hat{U}_{x,l})|^2). \tag{78}$$

Now, the desired result would follow upon invoking the inequality (75), once we show that $\omega([\hat{H}, \hat{U}_{x,l}]) = 0$. For any local $\hat{A}$ and real $t$, applying (69) to $\hat{V} = \hat{1} + t\hat{A}$ gives

$$t\,\omega([\hat{H}, \hat{A}]) + t^2 \omega(\hat{A}^\dagger[\hat{H}, \hat{A}]) \geq 0. \tag{79}$$

Since this holds for both signs of $t$, we get $\omega([\hat{H}, \hat{A}]) = 0$. □

## 4.2 Generalized LSM theorem

Let $\omega$ be a locally unique gapped ground state with energy gap $\Delta E > 0$ of a U(1) invariant Hamiltonian $\hat{H}$ on $\mathbb{Z}$. We denote by $\mathcal{T}_p$ the shift operator

$$\mathcal{T}_p(\hat{S}_j^{(\alpha)}) = \hat{S}_{j+p}^{(\alpha)}, \quad \alpha = x, y, z, \quad j \in \mathbb{Z}. \tag{80}$$

It is a linear $*$-automorphism: $\mathcal{T}_p(\hat{A}\hat{B}) = \mathcal{T}_p(\hat{A})\mathcal{T}_p(\hat{B})$ and $\mathcal{T}_p(\hat{A}^\dagger) = \mathcal{T}_p(\hat{A})^\dagger$. Suppose that $\omega$ is invariant under translation by $p$ in the sense that

$$\omega(\mathcal{T}_p(\hat{A})) = \omega(\hat{A}). \tag{81}$$

For any $l$ larger than $l_1 = \max\{C/\Delta E, l_0\}$, Lemma 4.8 guarantees that $\omega(\hat{U}_{x,l}) \neq 0$ for all $x$. Since the assumption (81) implies that

$$\omega(\hat{U}_{0,l}) = \omega(\hat{U}_{p,l}), \tag{82}$$

the mapping $[0, p] \ni x \mapsto \omega(\hat{U}_{x,l})$ then defines a closed loop in $\mathbb{C}\backslash\{0\}$. We denote the winding number of this path by $n_\omega$.

For a ground state satisfying (81), we define its filling factor as

$$\nu_\omega = \omega\left(\sum_{j=0}^{p-1}(\hat{S}_j^{(z)} + S)\right). \tag{83}$$

Note that the local operator $\hat{S}_j^{(z)} + S$ has integral eigenvalues, but $\nu_\omega$ need not be an integer.

**Lemma 4.9.** *Let $\omega$ be a locally unique gapped ground state of a* U(1) *invariant Hamiltonian $\hat{H}$, and assume that $\omega$ is invariant under translation by $p$. Then*

$$n_\omega = \nu_\omega. \tag{84}$$

*Proof.* Let $0 \leq x \leq p$ and $l \geq p$. Compare the twist angles of $\hat{U}_{x,l}$ and $\hat{U}_{0,l}$ site by site, using $e^{-2\pi i(\hat{S}_j^{(z)}+S)} = 1$: they differ by $-2\pi x/l$ on the sites $0 \leq j < l$, except for at most $2(p+1)$ sites near the two ends of the interval, where the difference is at most $2\pi p/l$ in magnitude. Since all $\hat{S}_j^{(z)}$ commute, this gives

$$\hat{U}_{x,l} = e^{i2\pi x\hat{\nu}_{0,l}}\,\hat{U}_{0,l} + \hat{R}_{x,l}, \qquad \|\hat{R}_{x,l}\| \leq \frac{C'}{l}, \tag{85}$$

with a constant $C'$ depending only on $p$ and $S$ (e.g. $C' = 8\pi p(p+1)S$), with

$$\hat{\nu}_{a,l} := \frac{1}{l} \sum_{a \le j < a+l} (\hat{S}_j^{(z)} + S) \quad (0 \le a < p). \tag{86}$$

Note that in the limit $l \to \infty$, we have $l = pN + O(1)$ for some large $N$, and

$$\omega(\hat{\nu}_{a,l}) = \frac{1}{p} \sum_{a \le j < a+p} \omega(\hat{S}_j^{(z)} + S) + O(l^{-1}) = \frac{1}{p}\nu_\omega + O(l^{-1}), \tag{87}$$

where the sub-leading term can be bounded by $C/l$ with $C$ independent of $a$ because there are only finitely many values $a$ take. Using this, we want to establish

$$\omega(\hat{U}_{x,l}) \approx e^{i2\pi x \nu_\omega/p} \omega(\hat{U}_{0,l}) \tag{88}$$

in the limit $l \to \infty$, which would show that the winding number of the curve $[0,p] \ni x \mapsto \omega(\hat{U}_{x,l})$ is $\nu_\omega$. To this end, we note that (85) implies

$$\begin{aligned}
&|\omega(\hat{U}_{x,l}) - e^{i2\pi x\nu_\omega/p}\omega(\hat{U}_{0,l})| \\
\le &|\omega((e^{i2\pi x\hat{\nu}_{0,l}} - e^{i2\pi x\nu_\omega/p})\hat{U}_{0,l})| + \frac{C'}{l} \\
\le &\left[\omega((e^{i2\pi x\hat{\nu}_{0,l}} - e^{i2\pi x\nu_\omega/p})(e^{-i2\pi x\hat{\nu}_{0,l}} - e^{-i2\pi x\nu_\omega/p}))\omega(\hat{U}_{0,l}^\dagger \hat{U}_{0,l})\right]^{1/2} + \frac{C'}{l} && \text{(89a)} \\
= &\, 2\omega(\sin^2[\pi x(\hat{\nu}_{0,l} - \nu_\omega/p)])^{1/2} + \frac{C'}{l} \\
\le &\, 2\pi x\, \omega((\hat{\nu}_{0,l} - \nu_\omega/p)^2)^{1/2} + \frac{C'}{l} && \text{(89b)} \\
\le &\, 2\sqrt{2}\pi x \Big\{\omega\big([\hat{\nu}_{0,l} - \omega(\hat{\nu}_{0,l})]^2\big) + [\omega(\hat{\nu}_{0,l}) - \nu_\omega/p]^2\Big\}^{1/2} + \frac{C'}{l}. && \text{(89c)}
\end{aligned}$$

In (89a) we applied Cauchy-Schwarz inequality to the hermitian form $(\hat{A}, \hat{B}) \mapsto \omega(\hat{A}^\dagger \hat{B})$. (89b) is obtained by expanding $\sin^2(x)$ into an alternating series. (89c) is essentially the inequality $(A+B)^2 \le 2(A^2+B^2)$. As will be stated in the next lemma, the first summand in the bottom line is small due to the *cluster property* of $\omega$. The second summand was already shown to be small in (87). □

**Lemma 4.10.** *Every pure state $\omega$ has the cluster property: for fixed local operators $\hat{A}$ and $\hat{B}$, with $\hat{B}_y := \mathcal{T}_y(\hat{B})$,*

$$\lim_{|y|\to\infty} [\omega(\hat{A}\hat{B}_y) - \omega(\hat{A})\omega(\hat{B}_y)] = 0. \tag{90}$$

*Proof.* See [BR87] or [Str21]. □

In particular,

$$\omega\big([\hat{\nu}_{a,l} - \omega(\hat{\nu}_{a,l})]^2\big) = \frac{1}{l^2} \sum_{j,k=\lceil a \rceil}^{\lceil a \rceil + l - 1} \omega\big([\hat{S}_j^{(z)} - \omega(\hat{S}_j^{(z)})][\hat{S}_k^{(z)} - \omega(\hat{S}_k^{(z)})]\big) \tag{91}$$

can be divided into two summations, $\sum_{|j-k|\geq D}$ and $\sum_{|j-k|<D}$, where $D$ is such that $\omega(\cdots) < \epsilon$ whenever $|j-k| \geq D$. Such a $D$ independent of $j$ exists because, by the $p$-periodicity of $\omega$, the summand depends only on $j \bmod p$ and $k-j$. Then

$$\omega\big([\hat{\nu}_{a,l} - \omega(\hat{\nu}_{a,l})]^2\big) < \frac{1}{l^2}\Big(l^2\epsilon + O(l)DS^2\Big) < 2\epsilon \tag{92}$$

for sufficiently large $l$. Since $\epsilon$ is arbitrary, the proof of Lemma 4.9 is now completed. We have thus established the following.

**Theorem 4.11** (generalized LSM theorem)**.** *Suppose $\hat{H}$ is a* U(1) *invariant Hamiltonian and it has a locally unique gapped ground state $\omega$. Further assume that $\omega$ is invariant under translation by $p$. Then the filling factor $\nu_\omega$ must be an integer.*

**Corollary 4.12** (Affleck-Lieb)**.** *Let $S$ be a half-odd integer. Then the antiferromagnetic Heisenberg chain does not have a unique gapped ground state.*

*Proof.* Global uniqueness implies local uniqueness and invariance under translation by 1. It remains to check that the filling factor is non-integer. By uniqueness, $\omega$ is also invariant under the rotation by $\pi$ about the $x$ axis, which maps $\hat{S}_0^{(z)} \mapsto -\hat{S}_0^{(z)}$; hence $\omega(\hat{S}_0^{(z)}) = 0$. Its filling factor is

$$\nu_\omega = \omega\left(\hat{S}_0^{(z)} + S\right) = S \in \mathbb{Z} + \frac{1}{2}. \tag{93}$$

□

# References


[BR87] Ola Bratteli and Derek W. Robinson. *Operator Algebras and Quantum Statistical Mechanics 1*. Springer Berlin Heidelberg, 1987. ISBN: 9783662025208. DOI: 10.1007/978-3-662-02520-8.

[Fra13] Eduardo Fradkin. *Field Theories of Condensed Matter Physics*. Cambridge University Press, Feb. 2013. ISBN: 9781139015509. DOI: 10.1017/cbo9781139015509.

[Gre13] Gerhard Grensing. *Structural Aspects of Quantum Field Theory and Noncommutative Geometry: (In 2 Volumes)*. WORLD SCIENTIFIC, Jan. 2013. ISBN: 9789814472708. DOI: 10.1142/8771.

[LSM61] Elliott Lieb, Theodore Schultz, and Daniel Mattis. "Two soluble models of an antiferromagnetic chain". In: *Annals of Physics* 16.3 (1961), pp. 407–466. ISSN: 0003-4916. DOI: https://doi.org/10.1016/0003-4916(61)90115-4.

[Str21] Franco Strocchi. *Symmetry Breaking*. Springer Berlin Heidelberg, 2021. ISBN: 9783662621660. DOI: 10.1007/978-3-662-62166-0.

[Tas20] Hal Tasaki. *Physics and Mathematics of Quantum Many-Body Systems*. Springer International Publishing, 2020. ISBN: 9783030412654. DOI: 10.1007/978-3-030-41265-4.

[Tas22] Hal Tasaki. *The Lieb-Schultz-Mattis Theorem: A Topological Point of View*. 2022. arXiv: 2202.06243 [cond-mat.stat-mech]. URL: https://arxiv.org/abs/2202.06243.

[Ton17] David Tong. *Statistical Field Theory*. 2017. URL: https://www.damtp.cam.ac.uk/user/tong/sft.html.